

A Method of Adaptive Low-Latency Downlink Transmission on FAPI based 5G NR gNB

Seung-Que LEE, Moon-Sik LEE
Communication & Media Research Laboratory
ETRI
Daejeon, KOREA
{sqlee, moonsiklee}@etri.re.kr

Abstract - This paper proposes a method to reduce downlink transmission delay in 5G NR gNB using FAPI interface. Each delay factor that may occur in the L2/L3 - L1SW - L1HW section was analyzed, and a variable section was predicted among them. In this paper, operation structure and operation procedure that can adaptively shorten the transmission delay in a variable section are presented. The proposed method is expected to help the fast response time of the 5G NR gNB.

Keywords - 5G NR gNB; FAPI; Low Latency Service; Downlink Scheduling

I. INTRODUCTION

FAPI (Functional Application Platform Interface) is an interface standard between L2/L3 and L1 established by SCF (Small Cell Forum). It is the de facto standard for designing/implementing various gNBs such as macro and micro cell gNBs as well as small cell gNBs. The use of FAPI clarifies the division of functional roles between two layers, which has been unclear so far, and reduces the implementation dependencies on the interworking of the two layers as much as possible, enabling evolution through independent development of each layer. As the traversing the FAPI messages to each layer, there are various delays such as communication latency and layer-specific processing time. In consideration of this, in-advance scheduling and delayed transmission are performed, which leads to an increase in service response time.

In this paper, we propose a method that enables early scheduling and fast transmission to be attempted adaptively to delay changes. We analyzed various delay properties and found delay elements that can be shortened. An operation structure to utilize such a delay factor was proposed, and an operation procedure based on the structure was also proposed.

II. DELAY ANALYSIS

A. Delay of FAPI Over UDP/IP

When the L2/L3 layer and the L1 layer are implemented on different platforms, the nFAPI should generally be used for the interface. However, if the connection between the two one is point-to-point only, a similar purpose can be achieved by simply communicating with FAPI over UDP/IP.[1][2] Fig. 1 shows the delay and jitter situation when FAPI is transmitted through UDP/IP. When L_{UDPU} and L_{UDPD} are the average UDP network latency in the up/down direction, respectively, and J_{UDPU} and J_{UDPD} are the up/down average jitter, respectively, the network delay has a maximum value of delay, $D_{UDP} = L_{UDPU} + J_{UDPU} + L_{UDPD} + J_{UDPD}$. If the downlink processing time of the L2/L3, T_{L23D} is added to this, the delay time of the L2/L3 is following. $D_{L23D} = D_{UDP} + T_{L23D}$.

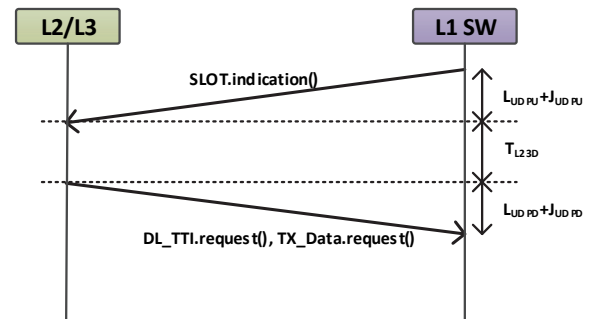


Fig. 1 Delay of FAPI Over UDP/IP

B. Delay of the L1 Control SW

The L1 Control SW is a functional block that controls the interface and modem between the upper layer and the physical layer. Here, there are three processing

procedures: reception & decoding, control register transfer, and MAC data transfer.[5] Fig. 2 shows the processing flow of the L1 control SW. Assuming that T_{DEC} , T_{PDUD} , and T_{MACD} are latency times for processing each function, and each procedure can be parallelized by other processors, the downlink delay of the L1 Control SW has the following values. $D_{LISWD} = T_{DEC} + \text{MAX}(T_{PDUD}, T_{MACD})$.

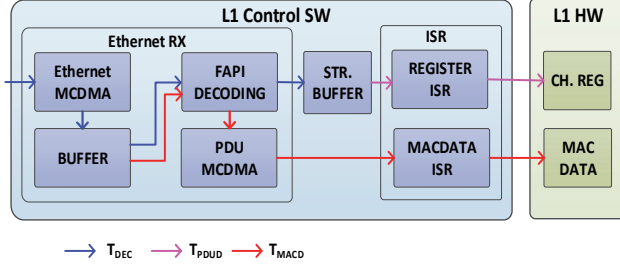


Fig. 2 Delay of the L1 Control SW

C. Delay of the L1 HW

The L1 HW goes through the steps of channel coding, modulation, resource mapping, precoding, IFFT as in fig.3.[3][4][6] Since each step is dependent on the previous step, the delay of L1 HW is generally the sum of the processing times of each function. So, the downlink modem delay is following. $D_{LIHWD} = T_{CHC} + T_{MOD} + T_{MAP} + T_{PRE} + T_{IFF}$. On the other hand, in a situation in which multiple CCs are processed, since each CC is processed in parallel by a separate HW, there is no change in the delay time.

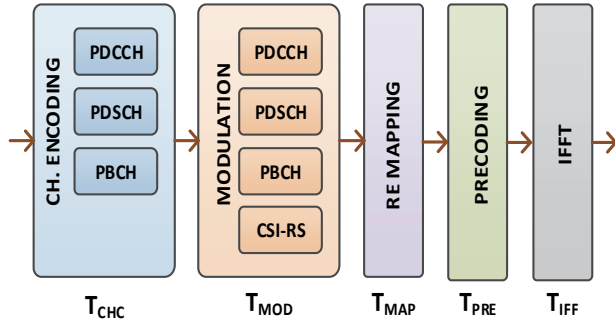


Fig. 3 Delay of the L1 HW

D. Downlink Scheduling Issue

Due to the various delay factors discussed above, the L2/L3 must make a transmission request ahead of the current slot time in downlink scheduling. For example, if a downlink signal is to be transmitted on the air interface at time t , the L2/L3 must make a transmission request in advance by at least $T_{DPRE} = D_{L23D} + D_{LISWD} + D_{LIHWD}$. If T_{DPRE} is as short as 1-2 slots, there is no problem. However, if this value is very large, response time is

increased and the quality of service is deteriorated. Therefore, it is necessary to find a way to improve it.

III. ADAPTIVE TRANSMISSION METHOD

A. Architecture

Fig. 4 shows the structures of the L2/L3, L1 SW, and the L1 HW of the 5G NR gNB proposed in this paper. Here, the sub-scheduler is provided in the L1 SW, which will be responsible for adjusting the schedule of the downlink PDCCH/PDSCH. In addition, the MAC scheduler and the sub-scheduler of the L1 SW have a structure in which schedule data is mutually shared. Therefore, the transmission schedule of the PDCCH/PDSCH determined by the MAC scheduler has a structure that can be changed by the sub-scheduler of the L1 SW.

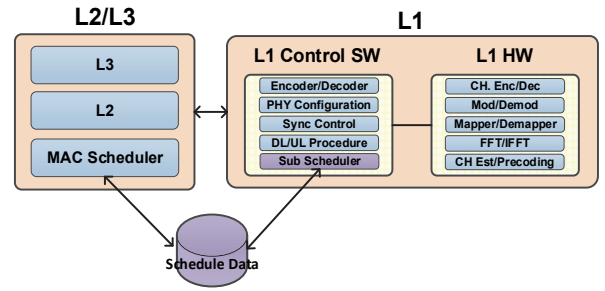


Fig. 4 Architecture

B. Procedure

Fig. 5 shows the method of adaptive low-latency downlink transmission proposed in this paper in relation to the FAPI messages and timing between L2/L3-L1 SW-L1HW. According to the delay analysis mentioned previously, the L2/L3 must transmit the downlink signal including DL-DCI-b in advance to L1 SW to SLOT $N - D_{L23D} - D_{LISWD} - D_{LIHWD}$ in order to transmit it on the air interface to SLOT N .

The L1 SW processes MAC PDU-a together with DL-DCI-b, and it is assumed that the required time is $D_{L23D} + D_{LISWD} - d1$ time (slot). If $d1$ is 0, it is processed in the same manner as in the conventional method. On the other hand, if $d1$ is greater than 0, since the processing time ends as early as the $d1$, the PDSCH/PDCCH transmission timing may be attempted earlier in the $d1$. However, if there is no radio resource blocks to transmit the PDSCH/PDCCH at that time, even in this case, the fast transmission is given up and the original method is performed.

Finally, if the condition for fast transmission is satisfied, the scheduling information of the PDSCH/

PDCCH is modified, and this event is notified to the L2/L3 layer. Then, after the downlink processing time D_{L1HWD} has elapsed, the MAC PDU-a of the PDSCH and the DL-DCI-b of the PDCCH are transmitted on the air interface in the slot $N-d1$. As a result, it has the effect that transmission is as fast as $d1$.

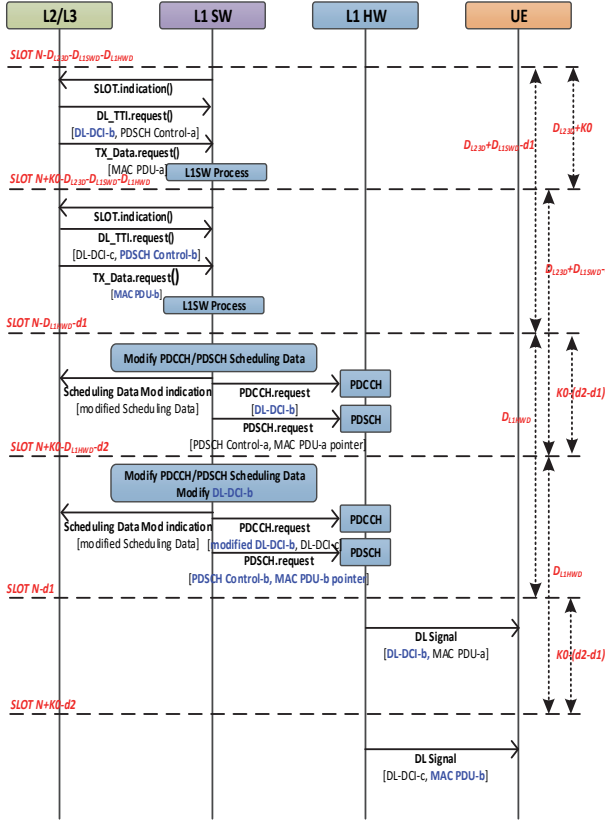


Fig. 5 Procedure

After that, the L2/L3 transmits the MAC PDU-b specified in DL-DCI-b to the L1 SW in advance to the $SLOT\ N+K0-D_{L23D}-D_{L1SWD}-D_{L1HWD}$ in order to transmit the MAC PDU-b to the slot $N+K0$ on the air interface.

Then, the L1 SW processes other DL-DCI-c together with MAC PDU-b, and it is assumed that the required time is $D_{L23D}+D_{L1SWD}-d2$ (slot). If $d2$ is 0, it is processed in the same way as in the original method. However, if $d2$ is greater than 0, since the processing time ends early in the $d2$, the transmission timing of the PDSCH is attempted to be transmitted in the $d2$ earlier. However, if there is no radio resource blocks to transmit the PDSCH/PDCCH at that time, the fast transmission is given up and the original method is proceeded. Otherwise, the L1 SW attempts to modify PDSCH scheduling information and PUCCH scheduling information for HARQ ACK/NACK reception among

PDCCH DL-DCI-b information related to MAC PDU-b for fast transmission. If the DL-DCI-b has already been transmitted on the air interface, the early transmission is abandoned and the original method is performed.

On the other hand, if it is determined that the time for sending DL-DCI-b is the same as the current slot or the L1 HW has already been requested, but it is not yet transmitted over the air interface, so a modification can be requested to the L1 HW for revised PDSCH/PDCCH scheduling information, and this event should be notified to the L2/L3 layer. After completing the modification of DL-DCI-b, the modified DL-DCI-b, PDSCH Control-b, MAC-PDU-b, and DL-DCI-c of other events request transmission to the L1 HW as quickly as $d2$. After the downlink processing time D_{L1HWD} has elapsed, it is transmitted on the air interface in the slot $N+K0-d2$. Therefore, transmission is possible faster than the originally planned $N+K0$ by $d2$.

IV. CONCLUSIONS

In this paper, we proposed a structure and procedure for adaptive scheduling and fast transmission to adapt variable delays. Since the L1 Control SW is implemented in software, the delay will be variable depending on the amount of work assigned to each slot, so it is expected that it will be effective in improving the overall response time if an early transmission is attempted adaptively.

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